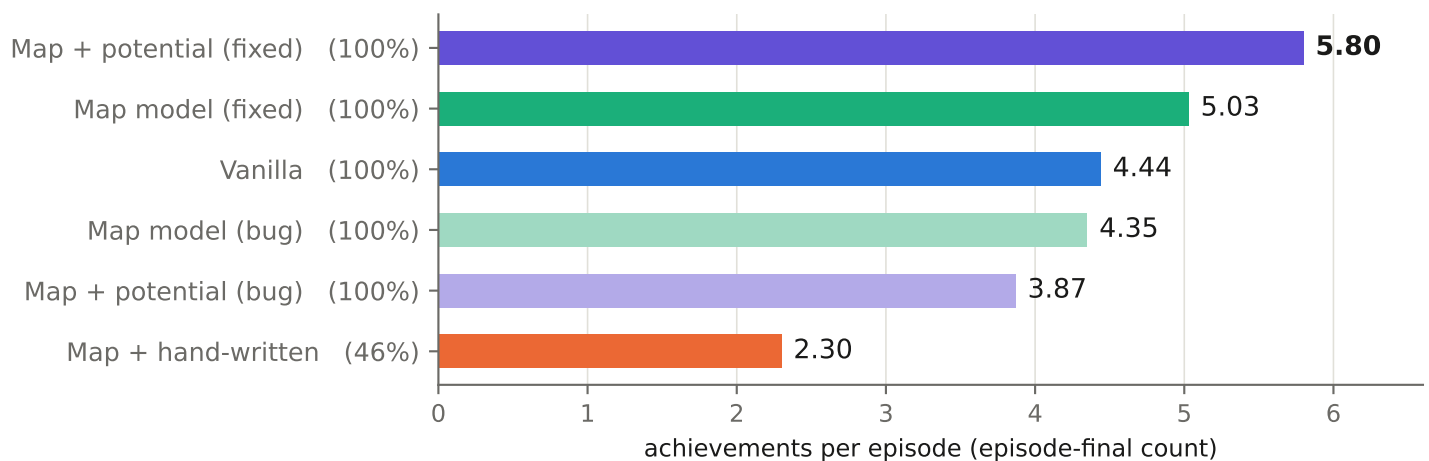


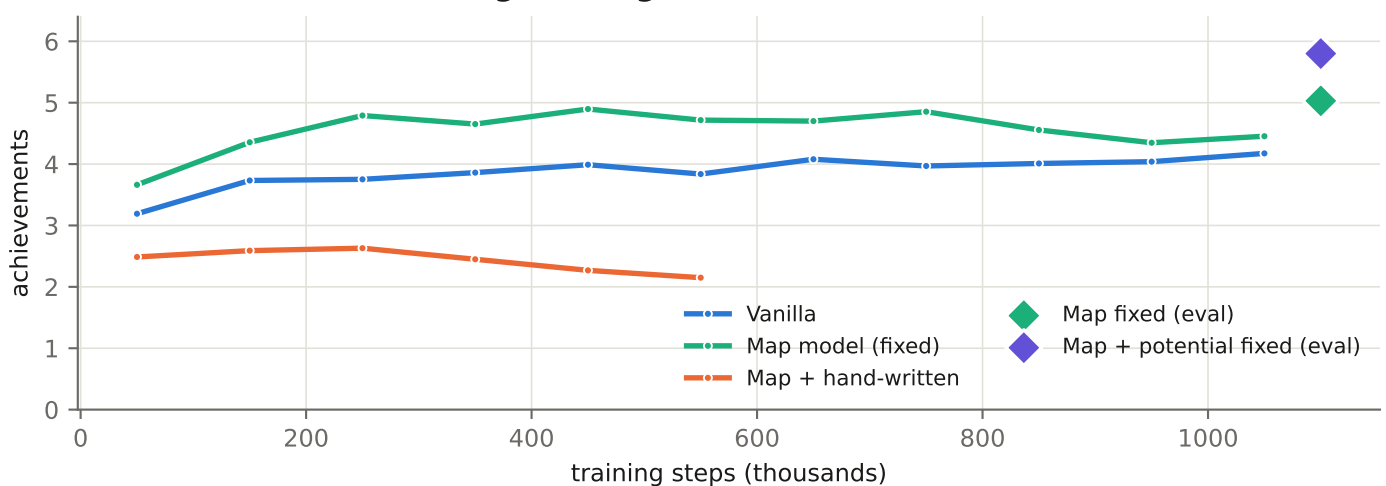
A one-flag bug fix, then potential shaping: 4.44 -> 5.03 -> 5.80 achievements

Achievement COUNT per episode · DreamerV3 · tools/death_eval.py, 25-30 episodes each

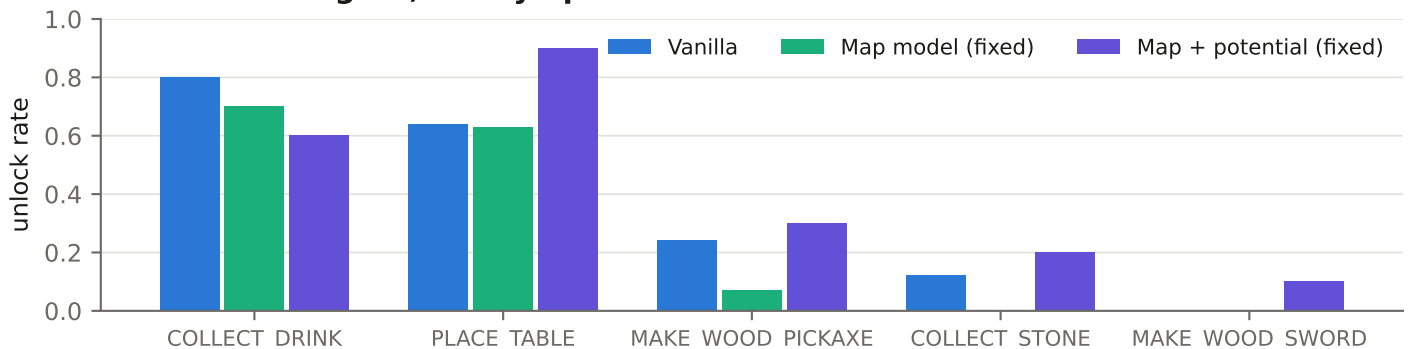
Achievement count at 1.1M steps, all runs



Achievement count during training



The tech-tree gate, finally opened



imag_shift True corrupted the policy gradient. The imagination rollout PICKS actions reading the crop frozen at its start, while the loss SCORES them reading the shifted crop - and imag_loss is REINFORCE, so it takes logpi of actions drawn from a different distribution. The objective ran to -61 where vanilla sat at 0.00. Turning the flag off is worth +0.68 on otherwise identical configs.

Potential-based shaping then adds +0.77 on top. They compose because they fix different things: the flag restores a working policy gradient, and PHI - weighted heaviest on MAKE_WOOD_PICKAXE - gives it somewhere to point. Under the bug the same shaping scored 3.87, below doing nothing at all.

The tech-tree gate opened. MAKE_WOOD_PICKAXE 0.30 and COLLECT_STONE 0.20 both beat vanilla; MAKE_WOOD_SWORD and PLACE_STONE appear for the first time in any run. Not fixed: episode length 278 sits inside the random-policy band and COLLECT_DRINK fell to 0.60. This bought depth, not survival.